Predicting Loan Default Using Supervised Learning: A Comparison of Logistic Regression and Random Forest

A Case Study Using Lending Club Data

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Abstract— **Assessing the creditworthiness of borrowers is a critical task for financial institutions seeking to minimise loan default risks. This project investigates the application of supervised machine learning to predict loan defaults using borrower financial and loan-related data from the Lending Club dataset. Two classification models, Logistic Regression and Random Forest, were implemented and compared. The dataset was preprocessed through feature selection, one-hot encoding, and standardisation of numerical variables. To address the significant class imbalance, balancing techniques including oversampling and undersampling were applied to enhance model sensitivity to defaults. Model performance was evaluated using accuracy, F1 score, ROC-AUC, and confusion matrices.**

The results show that the Random Forest model outperformed Logistic Regression model in detecting defaulting borrowers, achieving an F1 score of 0.868 and an ROC-AUC of 0.942. Additionally, 5-fold cross-validation confirmed the model's robustness, with consistent ROC-AUC scores across all folds. These findings demonstrate that customer financial history is a significant predictor of loan default and highlight the potential of interpretable machine learning models in credit risk assessment.

Keywords- Machine Learning, Credit Risk, Loan Default Prediction, Logistic Regression, Class Imbalance, Cross-Validation

I. INTRODUCTION

The increasing reliance on credit-based financial systems has made accurate credit risk assessment a critical function for lenders. Loan default, where borrowers fail to meet repayment obligations, poses significant financial risks to institutions, potentially leading to substantial losses. Traditional credit scoring models, while widely used, often rely on linear assumptions and limited variables, making them less effective in identifying complex patterns of borrower behaviour. As financial data grows in volume and complexity, there is a growing demand for more intelligent, data-driven methods of predicting credit risk.

This project addresses the problem of predicting loan default using historical financial data from Lending Club, an online peer-to-peer lending platform of the Lending Club Bank United States of America.

The central research question explored is: Can machine learning models based on customer financial history effectively predict the likelihood of loan default? This question is examined through the following hypotheses:

- Null Hypothesis (H_0): Customer financial history does not significantly predict loan default.
- Alternative Hypothesis (H_1): Customer financial history significantly predicts loan default.

This work aims to design and evaluate machine learning models, specifically Logistic Regression and Random Forest, that can classify loan applications as likely to default or be fully repaid. The study also explores how model performance is affected by imbalanced data and applies balancing techniques to improve the detection of rare default cases.

The key contributions of this project include:

- A full pipeline for credit risk prediction using supervised learning techniques.
- A comparison of two popular classification algorithms in the context of loan default.
- Evaluation of model performance using metrics such as accuracy, F1 score, and ROC-AUC.
- The use of 5-fold cross-validation to validate generalisability.
- A data-driven validation of the hypothesis that financial history predicts loan default.

The proposed solution leverages machine learning to learn patterns from borrower financial data and improve prediction accuracy compared to traditional methods. This approach is motivated by recent studies [2][3][4] which have shown the value of predictive modeling in credit scoring. By using accessible, interpretable, and scalable techniques, this project contributes a replicable framework that lenders or fintech platforms could adopt to enhance credit risk decision-making.

II. LITERATURE REVIEW

In recent years, the application of machine learning techniques to credit risk prediction has received growing attention. With access to large volumes of borrower data, traditional statistical models are being complemented or replaced by algorithms capable of identifying complex, non-linear patterns in financial behaviour.

Núñez Mora et al. [1] performed a comparative study on the performance of several machine learning models (including Logistic Regression, Random Forest, and XGBoost) using the LendingClub dataset. Their findings highlight Random Forest as a highly effective algorithm for binary classification, although Logistic Regression offered comparable results with significantly lower computational complexity. Additionally, the study addressed class imbalance using SMOTE and demonstrated that features such as loan amount, interest rate, and credit history were strong predictors of default.

Chang, Lee, and Sohn [2] examined the impact of feature engineering on model performance in binary classification problems, using a similar dataset. Their research showed that simple models like Logistic Regression could outperform more complex models when properly tuned and paired with relevant features. The study emphasised the importance of domain-specific feature selection and the role of regularisation in improving model generalisability.

Segun et al. [3] conducted a systematic review of 25 creditworthiness prediction studies and identified Logistic Regression, Random Forest, and Support Vector Machines as the most frequently adopted algorithms. The review also explored recent developments in hybrid models combining deep learning and traditional techniques, highlighting the need for balancing interpretability and predictive accuracy in financial applications.

From a foundational perspective, Hand and Henley [4] reviewed statistical classification methods commonly used in consumer credit scoring. The paper discussed issues such as reject inference, population drift, and the trade-offs between model transparency and performance. While Logistic Regression was praised for its interpretability and regulatory acceptance, the authors acknowledged the growing use of ensemble methods for their predictive power.

Together, these studies support the selection of Logistic Regression and Random Forest in this project. They also highlight common challenges such as data imbalance and feature importance, which this study addresses through class weighting, cross-validation, and performance comparison.

III. METHOD DESCRIPTION

A. The Dataset

This study uses a dataset available in Kaggle [5], a data science repository, for the 2007-2018 period, from Lending Club, a peer-to-peer lending platform that facilitates unsecured personal loans. The dataset contains detailed records of loan applications, including financial history, employment, income level, home ownership, and loan outcomes for each borrower. The version used in this project includes 848,454 loan records and 66 variables, including both numeric and one-hot encoded categorical features.

The dataset is well-structured and free from missing values. However, a significant class imbalance is present in the target variable loan_status, with the majority of loans being fully paid (class 0) and a minority being charged off or defaulted (class 1). To mitigate this imbalance, the dataset has been treated with oversampling and undersampling techniques.

The dataset includes key features such as:

No	Variable	Discription	Datatype
1	loan_amnt	Loan amount	float64
2	term	Loan repayment period	float64
3	sub_grade	Assigned score based on borrower's credit history	float64
4	emp_length	Borrow's employment length in years	float64
5	annual_inc	Borrower's annual income	float64
6	loan_status	Target variable (Defaulted of not)	float64
7	dti	Borrower's debt to income ratio	float64
8	mths_since_recent_inq	Months since most recent inquiry	float64
9	revol_util	Revolving line utilization rate	float64
10	num_op_rev_tl	Number of open revolving accounts	float64
11 - 16	home_ownership_	One-hot encoded 6 types of home ownerships	int64
17 - 66	addr_state_	One-hot encoded 50 states of addresses	int64

Table 1: Description and data types of variables

Outliers and inconsistent values were minimal due to prior preprocessing. Numerical features were standardised using the StandardScaler to ensure uniform scale across all variables. Variables that were already one-hot encoded in the dataset were retained as-is and integrated directly into the feature matrix.

To better understand relationships between variables, a correlation analysis was performed on numerical features. Features such as loan_amnt, revol_util, and dti showed moderate correlation with the loan status, while multicollinearity among predictors was generally low, supporting the use of models like Logistic Regression that assume independence among predictors.

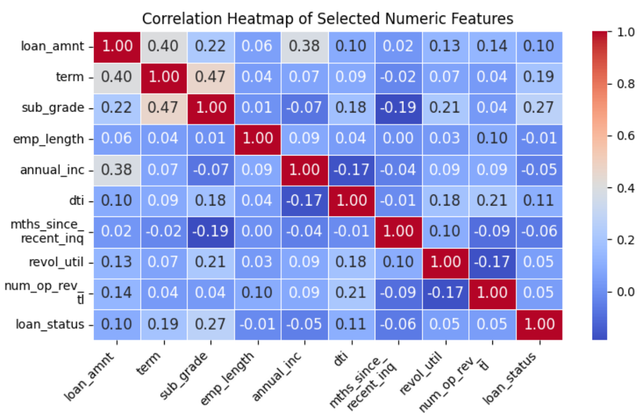

Fig 1: Correlation Heatmap of selected numerical features

A correlation heatmap of selected numeric features (Fig 1) showed that sub_grade had the strongest positive correlation with the target variable loan_status ($r = 0.27$), followed by term ($r = 0.19$), dti ($r = 0.11$), and loan_amnt ($r = 0.10$). These features showed moderate associations with loan outcomes, supporting their use in predictive models. In contrast, variables like annual_inc and emp_length had weak or no clear correlation with the target. Although Random Forest does not require independent features and can handle non-linear relationships, examining correlations helps in understanding how features relate to each other and the target. It also confirms that there is no strong overlap between predictors, which is particularly important for models such as Logistic Regression that assume low correlation among inputs.

The structure, completeness, and financial relevance of the dataset make it well-suited for developing predictive models to assess credit risk.

B. *Data Mining Method*

To solve the classification problem of predicting loan default, two supervised machine learning models were implemented: Logistic Regression and Random Forest. These models were selected based on their proven effectiveness in credit risk analysis tasks [1][2][3]. Logistic Regression is widely used in financial domains for its simplicity and interpretability [4], while Random Forest has been shown to perform well in complex, high-dimensional settings with imbalanced data [1], [2].

1) Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability of a binary outcome, in this case, whether a borrower will default. The model was trained using the liblinear solver with a maximum of 1000 iterations to ensure convergence. To address the significant class imbalance in the target variable, three different techniques were applied and compared:

- SMOTE (Synthetic Minority Over-sampling Technique) was used to generate synthetic samples of the minority class (defaults), increasing the number of default cases from 140,862 to 537,901.
- Random Under Sampling (RUS) reduced the number of majority class cases (non-defaults) to match the minority class, resulting in a balanced dataset with 140,862 records in each class.

These approaches were used independently to assess their effect on model performance. Prior research confirms that Logistic Regression performs competitively on Lending Club data and remains a widely accepted benchmark for credit scoring due to its simplicity and interpretability [1], [4].

2) Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their outputs to improve predictive accuracy. It is robust to overfitting and handles large feature sets well. In this project, a Random Forest Classifier was trained using 100 decision trees (n estimators=100) and class weight='balanced' to account for the skewed class distribution. Due to the large dataset size (over 848,000 records), training was performed on a random sample of 100,000 observations to reduce computation time and avoid memory issues. While this sampling approach improved model efficiency, it may have affected the generalisability and recall performance of the model compared to training on the full dataset. Random Forest has been widely recommended for classification problems involving consumer credit data [2], [3].

3) Data Processing and Model Invocation

The dataset was divided into features (X) and the target variable (y). Before training, numeric features were standardised using StandardScaler, and the model was trained on an 80% training set using train_test_split with stratification to preserve class proportions. The trained model was then used to predict outcomes on the remaining 20% test set. For each model, predictions were generated using predict () for class labels and .predict_proba() for probability estimates used in ROC-AUC evaluation.

4) Model Validation

To validate model performance and ensure generalisability, 5-fold cross-validation was conducted using the cross_val_score() function from scikit-learn. The scoring metric used was ROC-AUC, which assesses the model's ability to distinguish between default and non-default cases across different subsets of the data. The Logistic Regression model, in particular, demonstrated consistent ROC-AUC scores across all folds, confirming its robustness, a result supported by similar findings in prior studies [1], [3].

IV. RESULTS AND ANALYSIS

To evaluate the performance of the selected models in predicting loan default, both Logistic Regression and Random Forest were trained and tested on the processed Lending Club dataset. Their effectiveness was assessed using several evaluation metrics, including accuracy, precision, recall, F1 score, and the area under the receiver operating characteristic curve (ROC-AUC). To evaluate model performance, several classification metrics were used: accuracy, precision, recall, F1 score, and the area under the receiver operating characteristic curve (ROC-AUC). Accuracy reflects the proportion of total correct predictions. Precision refers to how many of the predicted defaults were actually correct. Recall measures how many of the actual defaults the model successfully identified. The F1 score provides a balance between precision and recall, especially useful in imbalanced settings. Finally, ROC-AUC is a measure of the model's ability to distinguish between default and non-default cases across various threshold values, where a value closer to 1 indicates better separation.

Since the dataset is highly imbalanced, with far fewer default cases compared to non-defaults, emphasis was placed on recall, F1 score, and ROC-AUC. These metrics provide a clearer understanding of how well the models detect the minority class, which is more important in the context of loan risk prediction than simply achieving high overall accuracy.

As summarised in Tables 1 and 2, the accuracy was low in the model trained using balanced data. However, the recall value of the model has increased drastically with the balanced data. Similarly, the F1 score and ROC-AUC have also increased, suggesting that the model can successfully detect the default cases using historical financial and loan-related data.

According to Tables 2 and 3, the highest accuracy of 79.1% was achieved by the model trained on the original imbalanced dataset. However, this model performed poorly in terms of recall (8.8%), F1 score (0.149), and precision, which remained below 50%. In contrast, the logistic regression (LR) model trained on the undersampled data showed a lower accuracy of 65.0% but achieved comparatively better precision and recall. Notably, the LR model trained on the oversampled dataset delivered the most balanced performance, with an accuracy of 68.0%, precision of 70.2%, and recall of 62.6%.

Metric	**Accuracy**	**Precision**	**Recall**
Imbalanced Training data	79.1%	48.4%	8.8%
Over-sampled training data with SMOTE	68.0%	70.2%	62.6%
Under-sampled Training data with RUS	65.0%	65.8%	62.7%

Table 2: Comparison of evaluation metrics for the Logistic Regression model under different class balancing techniques.

Metric	F1 Score	ROC-AUC
Imbalanced Training data	0.149	0.675
Over-sampled training data with SMOTE	0.662	0.757
Under-sampled Training data with RUS	0.642	0.708

Table 3: Comparison of F1 Score and ROC-AUC for Logistic Regression under different class balancing techniques.

The two confusion matrices (Fig. 2 and Fig. 3) of the two balancing methods show a good balance between correctly predicted defaulters and non-defaulters.

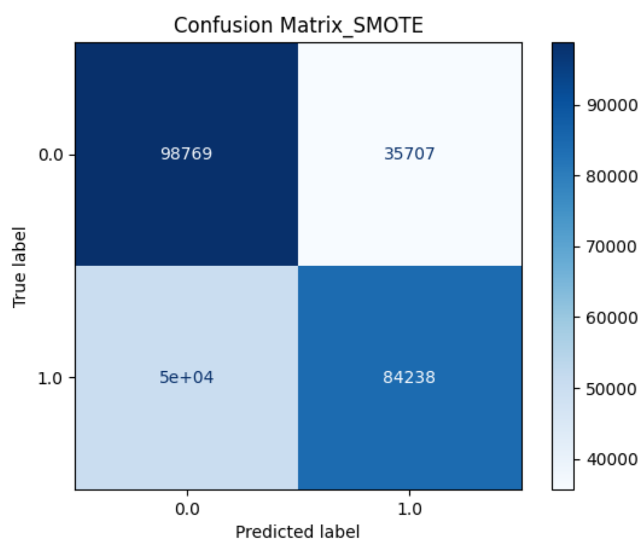

Fig 2: Logistic Regression Confusion Matrix (SMOTE)

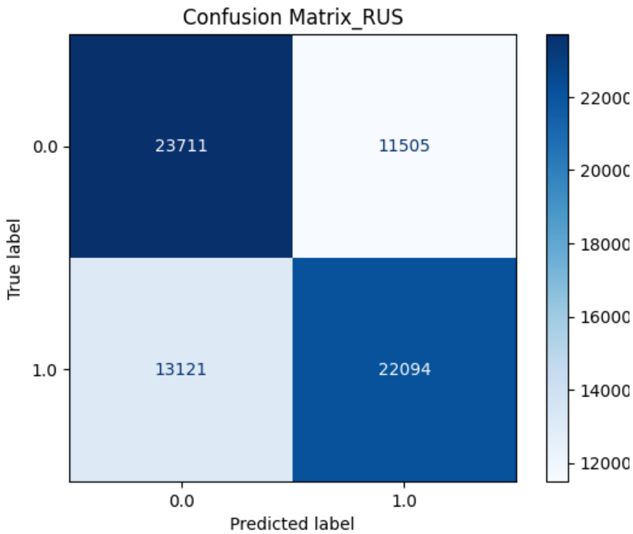

Fig 3: Logistic Regression Confusion Matrix (RUS)

The results of 5-fold cross-validation (Fig. 4) demonstrate consistency, with ROC-AUC values ranging from 0.706 to 0.709 and an average of 0.708. The mean accuracy of the 5-Fold cross validation method is 79.05%, and the standard deviation is 0.0008 shows that the logistic regression model is very consistent and has minimal variation across the folds. This indicates stable performance across different data splits.

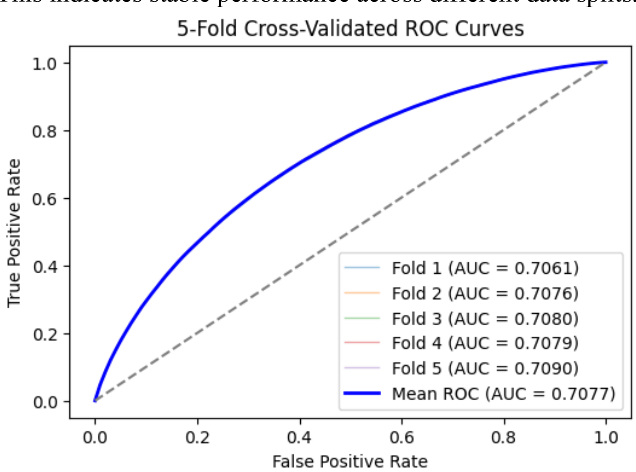

Fig 4: Logistic Regression 10-Fold Cross Validation ROC Curves

Metric	Random Forest SMOTE	Random Forest_RUS
Accuracy	87.4%	64.6%
Precision	90.8%	64.4%
Recall	83.2%	65.2%
F1 Score	0.868	0.648
ROC-AUC	0.942	0.701

Table 4: Comparison of Evaluation Matrices of RF Model

In contrast, the Random Forest model (RF), trained on a 100,000-sample subset due to computational constraints, uses two data balancing techniques, SMOTE and RUS. RF model trained with oversampled data achieved the highest overall accuracy of 87.4% compared to the accuracy of 64.6% of the model trained with undersampled data. The RF model trained with over-sampled data has identified 83.2% of actual default cases (True Positives). The precision value of the SMOTE model is 90.8%, while F1 score and ROC-AUC values show 0.868 and 0.942, respectively.

The confusion matrices of the two Random Forest models (Figures 5 and 6) indicate that both models achieve a reasonable balance in classifying both classes. However, the model trained with oversampled data demonstrates a stronger ability to correctly identify true negatives, while the model trained with undersampled data performs better in identifying true positives.

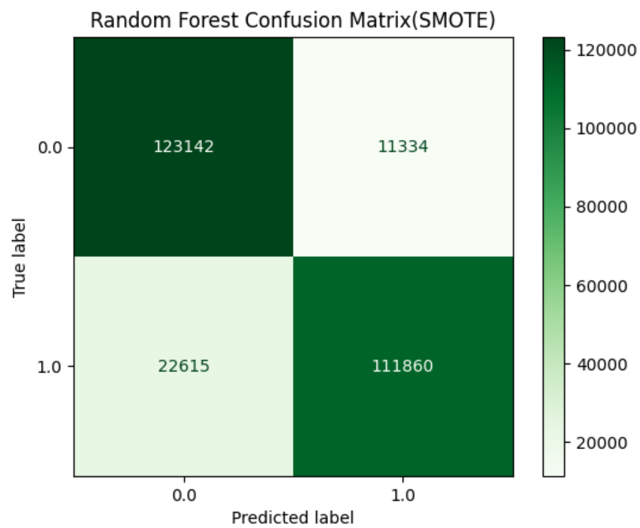

Fig 5: Random Forest Confusion Matrix (SMOTE)

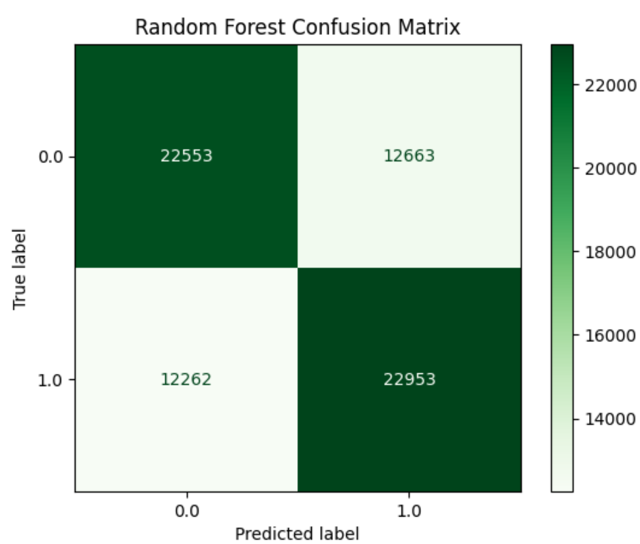

Fig 6: Random Forest Confusion Matrix (RUS)

Metric	Logistic Regression SMOTE	Random Forest SMOTE
Accuracy	68.0%	87.7%
Precision	70.2%	90.8%
Recall	62.2%	83.2%
F1 Score	0.662	0.868
ROC-AUC	0.757	0.942

Table 5: Comparison of evaluation metrics for Logistic Regression and Random Forest models under the SMOTE method.

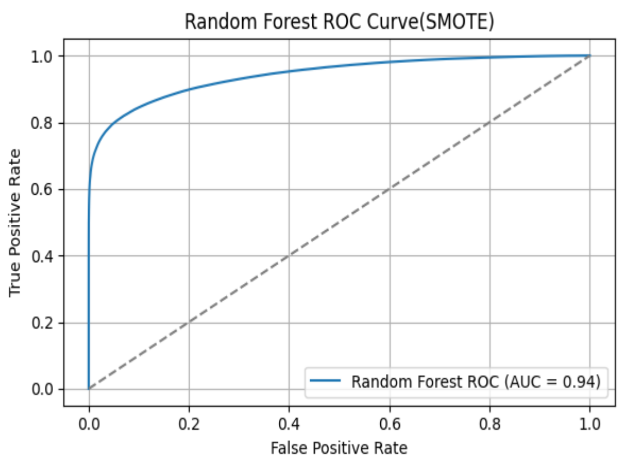

Fig 7: Random Forrest model ROC Curve (SMOTE)

As summarized in Table 5, the Random Forest model outperformed the Logistic Regression model across all evaluation metrics, including recall and F1 score, both of which are particularly crucial when dealing with imbalanced datasets. While Logistic Regression achieved an overall accuracy of 68.0% and a precision of 70.2%, it struggled to detect defaulters effectively, with a recall of only 62.2%. In credit risk applications, failing to identify a true default can be significantly more costly than a false positive. In contrast, the Random Forest model achieved a substantially higher accuracy of 87.7% and precision of 90.8%, and it managed the precision-recall trade-off more effectively, identifying more true defaulters with a recall of 83.2%. These results suggest that Random Forest is a more suitable choice for this classification task.

These findings are consistent with previous studies. Núñez Mora et al. [1] observed that Random Forest, as a highly effective algorithm for binary classification, although Logistic Regression offered comparable results with significantly lower computational complexity. It also highlighted Random Forest's limitations when applied to imbalanced data without resampling or specialised tuning techniques.

The ability of the Random Forest model to effectively use financial history variables, such as sub-grade, term, debt-to-income ratio, and loan amount, to detect defaults supports the rejection of the null hypothesis. The results confirm that customer financial history does significantly predict loan default.

V. Conclusion

This study aimed to investigate whether customer financial history can be used to predict loan default using supervised machine learning techniques. Two classification models, Logistic Regression and Random Forest, were implemented and evaluated on a real-world dataset from Lending Club. The Random Forest model demonstrated a superior ability to detect defaulters, as evidenced by its higher recall, precision, and F1 score. Additionally, it achieved a strong ROC-AUC score of approximately 0.94, indicating excellent overall classification performance. The results confirm that financial history features such as sub-grade, term length, and debt-to-income ratio are significant predictors of default. This supports the rejection of the null hypothesis and highlights the value of data-driven approaches in credit risk management.

This research contributes a replicable and interpretable modelling approach that could be refined further using additional financial behaviour data or by integrating advanced techniques such as ensemble methods or cost-sensitive learning. Future work may also explore incorporating time-series repayment behaviour and explainable AI techniques to enhance both predictive power and transparency.

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